

Lab 06

Due March 15, 5:00 PM PST

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Modify this template file to create your submission.

Problem 1a

```
## write your answer here
library(carData)
data("Prestige")
data_subset <- Prestige[, c("education", "income")]

set.seed(100)

prop_train <- 0.75
prop_test <- 1 - prop_train

n_data <- nrow(data_subset)

train_test <- sample(c("Train", "Test"),
                    size = n_data, replace = TRUE,
                    prob = c(prop_train, prop_test))

training_set <- data_subset[train_test == "Train",]
test_set <- data_subset[train_test == "Test",]
```

Problem 1b

```
## write answer here
fit_train <- lm(income ~ education, data = training_set)
fit_predictions <- predict(fit_train, newdata = test_set)

RMSPE <- sqrt(mean((fit_predictions - test_set$income)^2))
cat("RMSPE = ", RMSPE, "\n")
```

RMSPE = 2593.794

Problem 1c

```
## write answer here
library(caret)

K_vals <- seq(from = 1, to = 50, by = 1)
K_RMSPE <- rep(NA, length(K_vals))
set.seed(100)

CV_folds <- sample(1:5, size = nrow(training_set), replace = TRUE)
num_folds <- 5

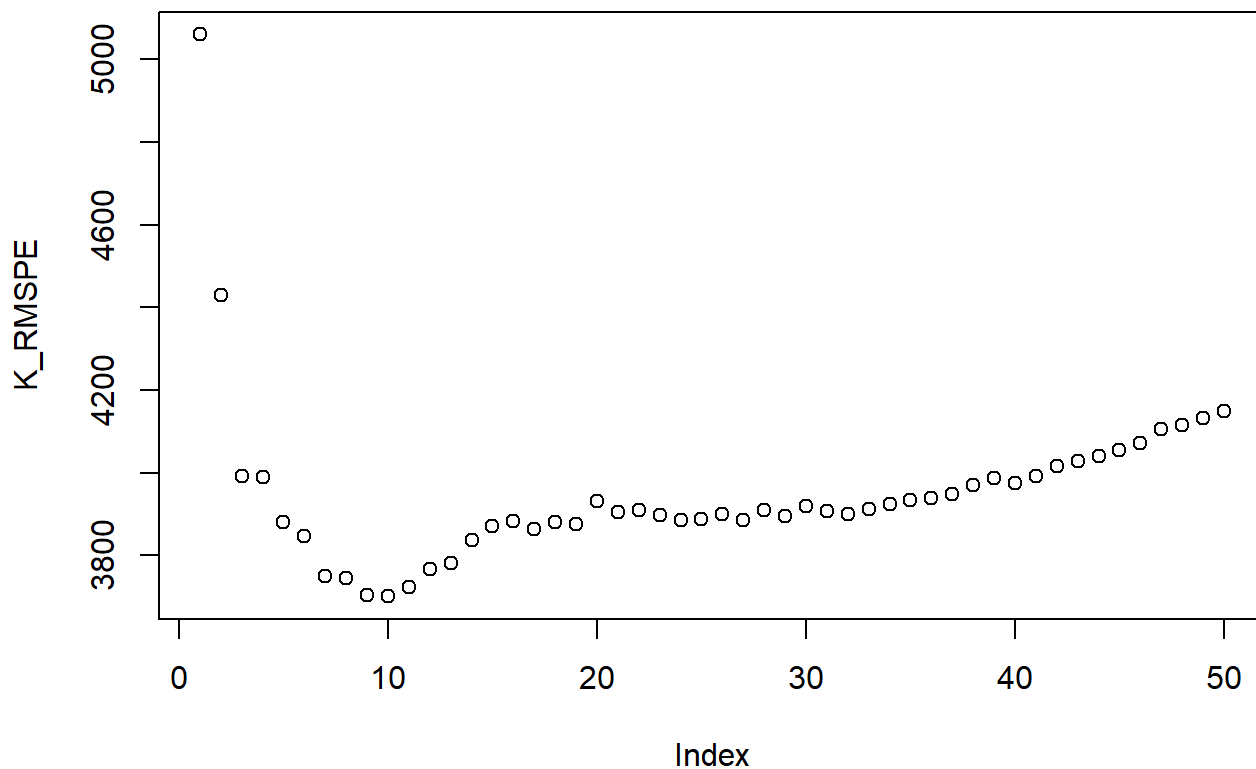
for (j in 1:length(K_vals)) {
  k <- K_vals[j]
  CV_RMSPE <- rep(NA, num_folds)

  for (i in 1:num_folds) {
    cv_train <- training_set[CV_folds != i,]
    cv_val <- training_set[CV_folds == i,]
    fit_knn <- knnreg(income ~ education, data = cv_train, k = k)
    cv_pred <- predict(fit_knn, cv_val)
    CV_RMSPE[i] <- sqrt( mean ((cv_pred - cv_val$income)^2))
  }
  K_RMSPE[j] <- mean(CV_RMSPE)
}

best_k <- K_vals[which.min(K_RMSPE)]
cat("Best K from cross-validation:", best_k, "\n")
```

Best K from cross-validation: 10

```
plot(K_RMSPE)
```



Problem 1d

write your answer here

```
fit_knn <- knnreg(income ~ education, data = training_set, k = 10)
fit_knn_prediction <- predict(fit_knn, test_set)
RMSPE_knn <- sqrt(mean((fit_knn_prediction - test_set$income)^2))
cat("RMSPE = ", RMSPE_knn, "\n")
```

RMSPE = 2189.479

```
cat("Yes, the knn model has a lower RMSPE value than linear regression, therefore it gives better
```

Yes, the knn model has a lower RMSPE value than linear regression, therefore it gives better predictions.